

RAG-Augmented Multi-Agent LLM Framework for Explainable Software Vulnerability Detection Using CWE Knowledge Bases

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One shape for every answer

This report adds a fixed answer form. It continues report 4.

What we did

Every answer must use the same fields. If a field is missing, the answer is rejected.

The fields are: the decision, the CWE, the cited lines, the cause, a short explanation, and a fix.

The decision can be unsafe, safe, or not sure.

The form has no place for attack steps. The tool is not asked to write an exploit.

We keep two sample answers in the project, one unsafe and one safe, so we can check the form before any model is called.

Line numbers start at 1. That is the same way a person counts lines in an editor.

If two runs use different shapes, we cannot put them in one table. The fixed form lets us compare the simple checker, the template, and the live model later.

The check is automatic. A person does not have to open every file to see if a field is missing.

Picture

Every answer uses the same fields

Decision

unsafe / safe / not sure

CWE

id and name

Lines

file and line numbers

Why

one short cause

Explain

plain words

Fix

what to change

Numbers for this step only

Field	What it means
Decision	Unsafe, safe, or not sure.
CWE	Id and name from the book.
Lines	The file and the line numbers we point at.
Fix	What a programmer should change.

What is next

Next we will run the steps in order: mark lines, search, write, then check the form.

What this report does not claim

The simple checker still only marks lines. It is not the final yes or no.